

An Early Developmental Marker of Deficit versus Nondeficit Schizophrenia

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Abstract

Background: People with schizophrenia who have primary negative symptoms (deficit schizophrenia) differ from those without such symptoms (nondeficit schizophrenia) relative to risk factors, course of illness, other signs and symptoms, treatment response, and biological correlates. Given the evidence for gestational and perinatal risk factors for schizophrenia, the deficit/nondeficit differences suggest these two groups have developmental differences as well. We previously reported a difference between people with schizophrenia and comparison subjects in the width of the palate. The palate attains its postnatal shape by gestational week 17. We tested the hypothesis that people with deficit and nondeficit schizophrenia would differ on palate width.

Methods: A dentist made blind assessments of the width, depth, and length of the palate in deficit (N=21) and nondeficit (N=25) patients and control subjects (N=127) matched for age and gender.

Results: Deficit patients had wider palates than both nondeficit patients ($P<0.0001$) and control subjects ($p<0.0001$), with respective effect sizes of 1.08 and 1.01. There were no significant deficit/nondeficit differences in palate length or depth, and no difference between nondeficit and comparison subjects in width, length, or depth.

Conclusions: Palate width reflects a divergence in development between deficit and nondeficit patients, which may occur by the early second trimester in some deficit patients. This difference is consistent with the hypothesis that deficit schizophrenia is a separate disease within the syndrome of schizophrenia. It may also point to a risk factor specific to deficit rather than nondeficit schizophrenia.

Key words:

pregnancy

negative symptoms

palate

delayed effects, prenatal exposure (MESH term)

cytomegalovirus

toxoplasmosis

1. Introduction

Schizophrenia has a great deal of heterogeneity in risk factors, signs and symptoms, course of illness, etiological/risk factors, and treatment response. This variation raises the question of whether schizophrenia is a syndrome rather than a single disease, analogous to, for instance, dementia, which can be caused by more than one disease.

One approach to understanding this heterogeneity has been study of people with primary negative symptoms. People with schizophrenia who have these features meet criteria for deficit schizophrenia, while people without these features have nondeficit schizophrenia (Kirkpatrick et al., 1989). As the deficit and nondeficit groups differ relative to the dimensions used to distinguish diseases—signs and symptoms, risk factors, course of illness, etiological/risk factors, and treatment response—we proposed that deficit schizophrenia is a separate disease within the syndrome of schizophrenia (Kirkpatrick et al., 2001). Subsequent research has provided additional evidence supporting this concept (Kirkpatrick & Galderisi, 2008; Kirkpatrick et al., 2017). Given the extensive evidence on prenatal and childhood/adolescent risk factors for schizophrenia (Matheson et al., 2011; Cannon et al., 2002; Geddes et al., 1999), one could expect that there would be developmental differences between deficit and nondeficit schizophrenia. One developmental risk factor for deficit but not nondeficit schizophrenia—an increase in summer rather than winter birth—has been replicated several times (Messias et al., 2004).

Minor physical anomalies (MPA), which have an increased prevalence in schizophrenia, are theoretically interesting because the gestational time period in which each one develops is usually known (Compton et al., 2011; Franco et al., 2010). An abnormal palate is part of the Waldrop physical anomaly scale, which has been used extensively in study of MPAs in schizophrenia and other disorders, but little attention has been paid to this single item. That scale rates for the presence (abnormal) or absence of a high or steeped palate. A blinded study using casts of the palate and teeth found an increase in width but not depth or length of the palate in people with schizophrenia, with a large effect size (Kirkpatrick et al., 2007). A second blinded study (Delice et al., 2016) of palate casts from people with schizophrenia and comparison subjects also found an increase in palate width.

We hypothesized that an abnormally wide palate would be found in deficit but not nondeficit patients and tested this hypothesis in samples from the study of Delice et al. (2016).

2. Methods and Materials

2.1 Subjects:

Subjects and data came from the study of Delice et al. (2016), and further details of the sample and measurement methods can be found in the original report. In brief, they made casts of the palate of their subjects, then measured palate width, length, and depth blind to group membership (schizophrenia vs. comparison subjects) using landmarks as described (Delice et al., 2016). In that study, 56 people in that study who had schizophrenia had ratings on the Positive and Negative Syndromes Scale (PANSS) and were included in the present study, along with 127 comparison subjects without schizophrenia matched to the patients on age and gender. All of the subjects were Turkish.

2.2 Deficit/nondeficit categorization

We used the Proxy for the Deficit Syndrome to distinguish deficit vs. nondeficit patients, as used previously (for instance, Garcia-Rizo et al., 2012; Kirkpatrick et al., 1993; Messias et al., 2001). This approach has been validated by comparison categorizations based on the Schedule for the Deficit Syndrome, the standard instrument for the diagnosis of deficit and nondeficit patients (Kirkpatrick et al., 1989, 1993; Messias et al., 2001). The PDS quantifies the combination of high

negative symptoms and low emotionality (guilt, hostility, anxiety, and depression) that is characteristic of deficit compared to nondeficit subjects (Kirkpatrick et al., 1989, 1993, 1994). We calculated a continuous variable, PDS, using PANSS items scores as follows to quantify the combination of elevated negative symptoms and decreased emotionality/distress:

$$PDS = (n1+n4) - (p7+g2+g3+g6) = (\text{blunted affect} + \text{passive/apathetic social withdrawal}) - (\text{hostility} + \text{anxiety} + \text{guilt feelings} + \text{depression})$$

Scores ranged from 1 to -8. Patients with scores from -1 to 1 were assigned to the putative deficit group (N=21); patients with scores from -2 to -8 were assigned to the putative deficit group (N=35).

We then tested the validity of this categorization by comparing the clinical and demographic features of the two patient groups. A valid deficit group would have more severe negative symptoms but would not have greater hallucinations + delusions or disorganization. By definition the putative deficit group had decreased emotionality.

2.3 Analyses:

Separately for palate width, length and depth, we conducted analyses of variance, with the two patient groups and a comparison group without known psychotic symptoms--matched for age and gender—as the independent variable, and width, depth, or length of the palate as the dependent variable. We used Tukey's test for followup analyses.

3. Results

The three subject groups—deficit and nondeficit patients and comparison subjects—had very similar mean ages and male/female percentages. The deficit group had the expected clinical characteristics compared to nondeficit schizophrenia subjects: greater negative symptoms and decreased emotionality, but not greater psychotic symptoms (**Table 1**). Medication treatment would not alter palate dimensions, but the two groups did not differ with regard to percentages on first vs. second generation/drugfree/taking clozapine (data not shown).

The deficit group had a significantly wider palate than both the nondeficit patients and comparison groups but did not differ from them on palate length or depth (**Table 2**); the average difference as 3.8 mm. The effect size for the deficit/nondeficit difference was 1.08 (Cohen's *d*), and 1.01 for the deficit/comparison group difference. The power to detect a nondeficit/control difference of equal to that of the deficit/control difference (3.8 mm) was 0.99. The power to detect a nondeficit/control subject difference of 2.0 mm was 0.92, and of 1.5 mm was 0.71 (<https://homepage.divms.uiowa.edu/~rlenth/Power/index.html>).

4. Discussion

In a blinded study, people with features of deficit schizophrenia had a significantly wider palate than did people with nondeficit schizophrenia, with a large effect size for the deficit/nondeficit difference and the deficit/comparison subject difference. This finding is consistent with findings in another blinded study of an effect size of 0.80 compared to control subjects when the two patient groups were not separated (Kirkpatrick et al., 2007). This difference was not confounded by age, gender, or ethnicity. The two patient groups did not differ with regard to palate width or length, nor did the nondeficit group differ from the comparison subjects on any palate measures. The statistical power to detect the same sized difference in palate width between the nondeficit group and the control group, or a difference half the size of that of the deficit/control group difference was very strong, so the failure to find a difference between those two groups was unlikely to be due to a type II error.

We did not have information on socioeconomic status of the family of origin, which might confound the relationship between palate depth and clinical features. In addition, we cannot assess the specificity of palate depth relationship to deficit schizophrenia, that is, other prenatal markers

may also distinguish the two schizophrenia groups. Although the large effect sizes are encouraging, this finding would need to be replicated.

This abnormality in the deficit patients may reflect abnormal development during a specific period of gestation. Barring injury or other unusual events, a recognizable palate forms between weeks 6 and 17 (Diewert et al., 1993; Diewert & Wang, 1992). This is a period when most neurogenesis, and much of neuronal migration, takes place (Kumar, 2008); deficit patients may have an abnormality of early neuronal migration (Kirkpatrick et al., 1999, 2003). Our finding, then, suggests may be a divergence in development between people with deficit vs. nondeficit schizophrenia that occurs no later than the early second trimester, although the possibility of abnormal development in later life in some or all people with deficit schizophrenia cannot be ruled out by our data.

There are hints in the literature of what may underlie the deficit group's abnormally wide palate. We previously reported a greater prevalence of seropositivity for CMV, but not other human herpes viruses, in deficit compared to nondeficit patients (Dickerson et al., 2006). A significant association between CMV and cleft palate, intellectual disability, and/or deafness in children has also been found (Divya et al., 2017). As cleft lip/cleft palate is associated increased palate width (Manyama et al., 2014), congenital CMV may contribute to the group difference in palate shape that we found. In addition, some (Shirts et al., 2008; Watson et al., 2013; Houenou et al., 2014) but not all (Yolken et al., 2011; Dickerson et al., 2014) studies have found an association between CMV seropositivity and greater cognitive impairment within schizophrenia, and one study found negative symptoms were greater in patients with CMV seropositivity (Bolu et al., 2016). In view of the results in children, it is also of note that people with deficit schizophrenia have greater cognitive impairment than do those with nondeficit schizophrenia (Cohen et al., 2007). Seropositivity for CMV in a nonpatient sample has been associated with decreased novelty seeking (Novotna et al., 2005); whether such a change would increase negative symptom ratings is not known.

Taken together, this information raises the question of whether CMV might be a specific risk factor for deficit schizophrenia. However, a systematic review at the individual level of prenatal maternal CMV infection and risk of schizophrenia in the offspring did not find an association between infection and increased risk (Khandaker et al., 2013). Subsequent studies have suggested that there is a gene-environment interaction schizophrenia risk, relating CMV seropositivity and either of two DNA markers near the CTNNA3 gene on chromosome 10q21.3, and near D6S2672 in the HLA region (Borglum et al., 2014; Kim et al., 2007; Avramopoulos et al., 2015) and risk of schizophrenia. Perhaps CMV is associated with increased risk of schizophrenia only in people with particular DNA variants, and as a consequence an association with schizophrenia would only be found if the deficit group were considered separately. Studies with a design different from our present study would be needed to assess this issue.

Another possible risk factor for the palate difference we found is seropositivity to *Toxoplasma gondii*, although the suggestive evidence is weaker. There is some evidence toxoplasmosis seropositivity may also be associated with cleft palate (Erdelyi, 1957; Niebyl et al., 1985; Holková et al., 1994), and seropositivity is associated with an increased risk of schizophrenia (Sutterland et al., 2015). For toxoplasmosis to play a role in the abnormal development of the palate it is likely that it would usually have to affect the patient early in gestation. A systematic review of congenital toxoplasmosis yielded an incidence rate of 0.6 cases per 1000 births in North America (Torgerson & Mastroiacovo, 2013). *Toxoplasma* seropositivity does not appear to be associated with poorer cognition within schizophrenia (Shirts et al., 2008; Dickerson et al., 2014; Yolken et al., 2011). In a large population-based study (Sugden et al., 2017), seropositive subjects had poorer cognitive function than those who were seronegative in just one of fourteen cognitive

tests, a difference that might have been due to chance. The hypothesis that *Toxoplasma gondii* is associated with deficit compared to nondeficit schizophrenia has not been tested to our knowledge.

Another interpretation of the evidence on CMV and toxoplasmosis is that any adverse pregnant event, infectious or otherwise, could increase the risk of a wide palate and deficit schizophrenia if it occurred very early in gestation.

We are not aware of other data on deficit/nondeficit differences or similarities in other gestational measures of relevance to schizophrenia, such as obstetrical complications, exposure to prenatal stress, or birth weight (Brown, 2012; Cannon et al., 2002). Nor are we aware of any evidence on differences or similarities in the prevalence of other minor physical anomalies that are found in schizophrenia as a whole—which are due to abnormal prenatal development—in deficit and nondeficit schizophrenia.

Both the present finding and the associations of CMV with both cleft palate and deficit vs. nondeficit schizophrenia would need replication. If replicated the association of a wider palate would be a marker of a different developmental history for deficit compared to nondeficit schizophrenia, and would support the concept that deficit schizophrenia is a separate disease within the syndrome of schizophrenia (Kirkpatrick et al., 2001; Kirkpatrick and Galderisi, 2008; Kirkpatrick et al., 2017).

Declaration of Interests:

Dr. Kirkpatrick receives licensing royalties from ProPhase LLC for use of the Brief Negative Symptom Scale (BNSS) by for-profit groups; these fees are donated to the Brain and Behavior Research Foundation. He has also received honoraria and travel support from ProPhase LLC for training pharmaceutical company raters on the BNSS, consulting fees and travel support from Genentech/Roche, Minerva Neurosciences, and ProPhase LLC, consulting fees from anonymized pharmaceutical companies through Decision Resources, Inc., from an investment capital company through Guideposts, and fees for consulting on a legal issue involving Janssen Pharmaceutica and Wockhardt Bio. He also receives fees from Walsh Medical Media for editorial services and received fees for editorial services from Physicians Postgraduate Press, Inc.

Dr. Fernandez-Egea received consulting honoraria from Recordati and Angelini pharmaceuticals.

Table 1: Clinical and Demographic Characteristics in Deficit and Nondeficit Patients

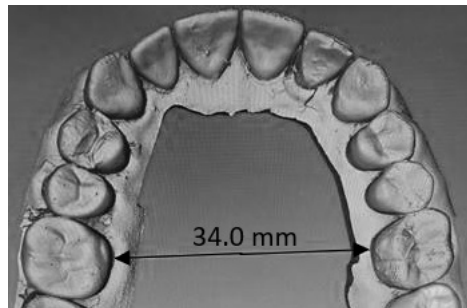
	Deficit (N=21)	Nondeficit (N=35)	Controls	p value
Age	34.2 (7.5)	34.5 (6.3)	34.1 (6/0)	0.95
% male	52%	48%	51.2	0.95
PANSS negative symptoms	15.6 (2.8)	11.7 (3.2)		>0.001
Hallucinations + delusions	2.6 (0.9)	3.2 (1.4)		0.10
Disorganization	1.4 (0.6)	1.6 (0.9)		0.21

Table 2: Palate Dimensions in Deficit and Nondeficit Patients, and Control Subjects

	Deficit (N=21)	Nondeficit (N=35)	Control Subjects (N=127)	F value	d.f.	p value
Mean width	37.5 (3.9)	33.7 (3.1)	34.0 (2.9)	15.37	2,186	> 0.001 *
Mean length	33.9 (2.2)	34.9 (3.2)	33.3 (3.2)	1.32	2,186	0.27
Mean depth	22.0 (2.4)	22.6 (2.2)	21.6 (2.2)	0.74	2,186	0.48

* Tukey posthoc test: deficit > nondeficit $p < 0.001$; deficit > control $p < 0.0001$; nondeficit not significantly different from control ($p < 0.83$).

Figure



Legend: "Palate width was measured with a digital caliper, and defined as the distance between a landmark on the two (right and left) upper (maxillary) first molars. The landmark used was the intersection of the free gingival margin and a vertical line dropped down from the mesiolingual cusp tip."